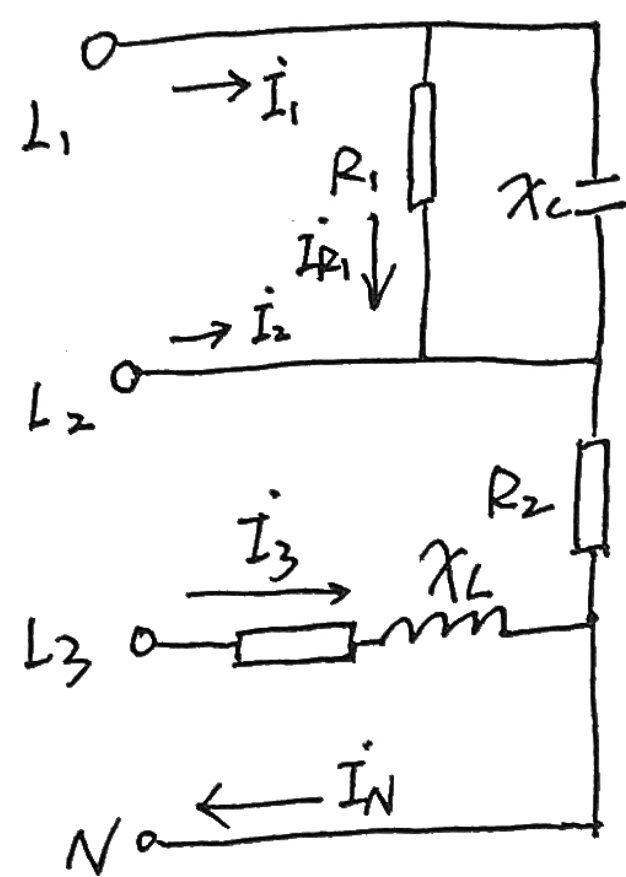




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图示三相电路中, 已知:  $\dot{U}_{12} = 380 \angle 0^\circ \text{ V}$   $R_1 = X_c = 38 \Omega$

$R_2 = 22 \Omega$   $R_3 = 10 \Omega$   $X_L = 10\sqrt{3} \Omega$

求: (1) 电流  $\dot{I}_1$ 、 $\dot{I}_2$ 、 $\dot{I}_3$ 、 $\dot{I}_N$ ;

(2) 电路无功功率。

解: (1) 由题可知:  $\dot{U}_{12} = 380 \angle 0^\circ \text{ V}$   $\dot{U}_{23} = 380 \angle -120^\circ \text{ V}$   $\dot{U}_{31} = 380 \angle 120^\circ \text{ V}$   
 $\dot{U}_1 = 220 \angle -30^\circ \text{ V}$   $\dot{U}_2 = 220 \angle -150^\circ \text{ V}$   $\dot{U}_3 = 220 \angle 90^\circ \text{ V}$

$$\dot{I}_1 = \dot{I}_{R1} + \dot{I}_{Xc} = \frac{\dot{U}_{12}}{R_1} + \frac{\dot{U}_{12}}{-jX_c} = 10\sqrt{2} \angle 45^\circ \text{ A}.$$

$$\dot{I}_2 = \dot{I}_{R2} - \dot{I}_1 = \frac{\dot{U}_2}{R_2} - \dot{I}_1 = 23.9 \angle -141.2^\circ \text{ A}$$

$$\dot{I}_3 = \frac{\dot{U}_3}{R_3 + jX_L} = 11 \angle 30^\circ \text{ A}.$$

$$\dot{I}_N = \dot{I}_3 + \dot{I}_{R2} = \dot{I}_3 + \frac{\dot{U}_2}{R_2} = 1 \angle 30^\circ \text{ A}$$

(2) 无功功率  $Q = I_3^2 X_L - \frac{U_{12}^2}{X_c} = 1210\sqrt{3} - 3800 \text{ var}$